Untitled66.ipynb

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[1]
```
# Matplotlib
# matplot library
# pyplot module
```

[2]
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

[3]
```
# for demo dataset use deaborn library
import seaborn as sns
```

[4]
```
df = sns.load_dataset("titanic")
```

[5]
```
df
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
886	0	2	male	27.0	0	0	13.0000	S	Second	man	True	NaN	Southampton	no	True
887	1	1	female	19.0	0	0	30.0000	S	First	woman	False	B	Southampton	yes	True
888	0	3	female	NaN	1	2	23.4500	S	Third	woman	False	NaN	Southampton	no	False
889	1	1	male	26.0	0	0	30.0000	C	First	man	True	C	Cherbourg	yes	True
890	0	3	male	32.0	0	0	7.7500	Q	Third	man	True	NaN	Queenstown	no	True

891 rows × 15 columns

Next steps: Generate code with df New interactive sheet

[6]
```
df.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Next steps: Generate code with df New interactive sheet

[7]
```
df.tail()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
886	0	2	male	27.0	0	0	13.00	S	Second	man	True	NaN	Southampton	no	True
887	1	1	female	19.0	0	0	30.00	S	First	woman	False	B	Southampton	yes	True
888	0	3	female	NaN	1	2	23.45	S	Third	woman	False	NaN	Southampton	no	False
889	1	1	male	26.0	0	0	30.00	C	First	man	True	C	Cherbourg	yes	True
890	0	3	male	32.0	0	0	7.75	Q	Third	man	True	NaN	Queenstown	no	True

[9]
```
df.tail(1)
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
890	0	3	male	32.0	0	0	7.75	Q	Third	man	True	NaN	Queenstown	no	True

[10]
```
# BAR Chart -
# Cateogrical - count -frequency
# excel - pivot
```

[11]
```
df["sex"].unique()
```

```
array(['male', 'female'], dtype=object)
```

[12]
```
df["sex"].value_counts()
```

sex	count
male	577
female	314

```
dtype: int64
```

[16]
```
gender = df["sex"].value_counts()
```

[17]
```
gender
```

sex	count
male	577
female	314

```
dtype: int64
```

[18]
```
gender.index
```

```
Index(['male', 'female'], dtype='object', name='sex')
```

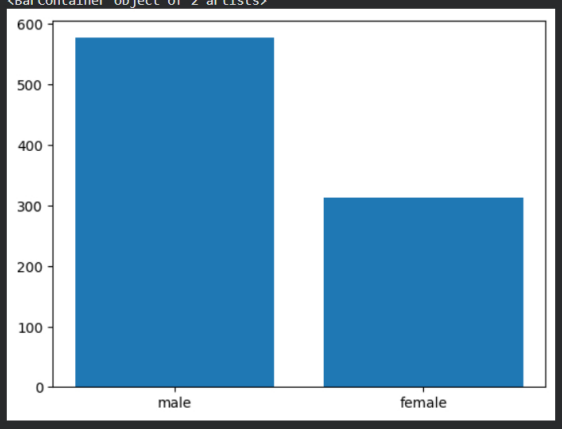
[19]
```
gender.values
```

```
array([577, 314])
```

[20]
```
# use bar chart from matplotlib

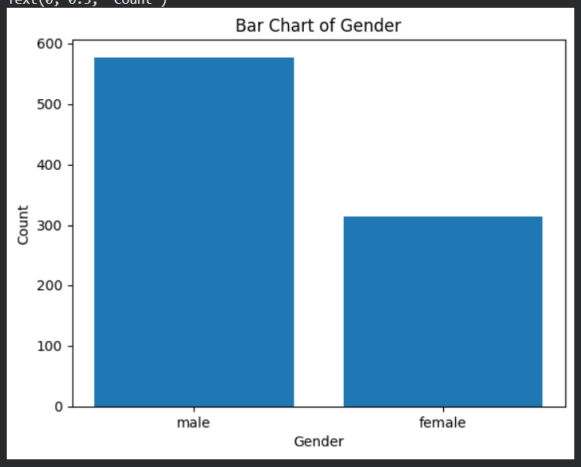
plt.bar(gender.index , gender.values)
```

```
<BarContainer object of 2 artists>
```



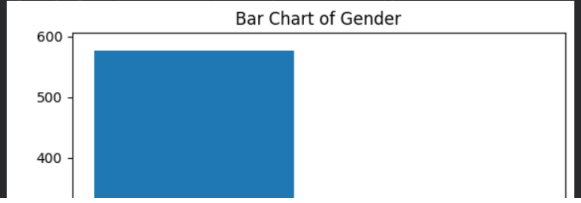
[21]
```
plt.bar(gender.index , gender.values)
plt.title("Bar Chart of Gender")
plt.xlabel("Gender")
plt.ylabel("Count")
```

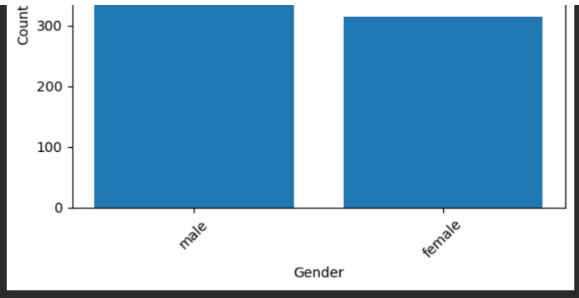
```
Text(0, 0.5, 'Count')
```



[24]
```
plt.bar(gender.index , gender.values)
plt.title("Bar Chart of Gender")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.xticks(rotation = 45)
```

```
([0, 1], [Text(0, 0, 'male'), Text(1, 0, 'female')])
```

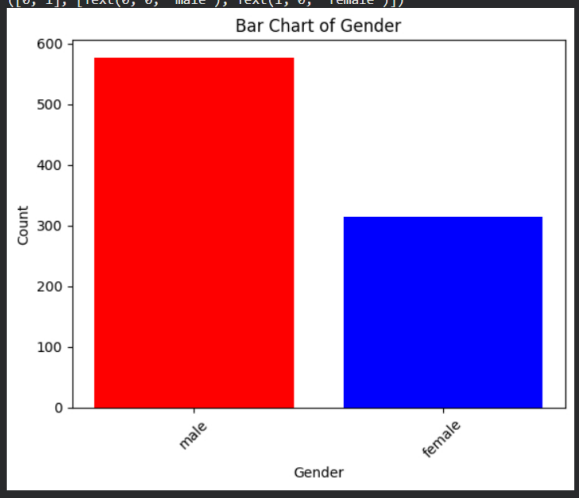




[25]
✓ 0s

```
plt.bar(gender.index , gender.values , color=["red", "blue"])
plt.title("Bar Chart of Gender")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.xticks(rotation = 45)
```

```
([0, 1], [Text(0, 0, 'male'), Text(1, 0, 'female')])
```



[26]
✓ 0s

```
gender
```

sex	count
male	577
female	314

dtype: int64

[27]
✓ 0s

```
gender1 = gender.reset_index()
```

[29]
✓ 0s

```
gender1   #dataframe
```

	sex	count
0	male	577
1	female	314

Next steps: Generate code with gender1 | New interactive sheet

[30]
✓ 0s

```
gender1["sex"]
```

	sex
0	male
1	female

dtype: object

[31]
✓ 0s

```
gender1["count"]
```

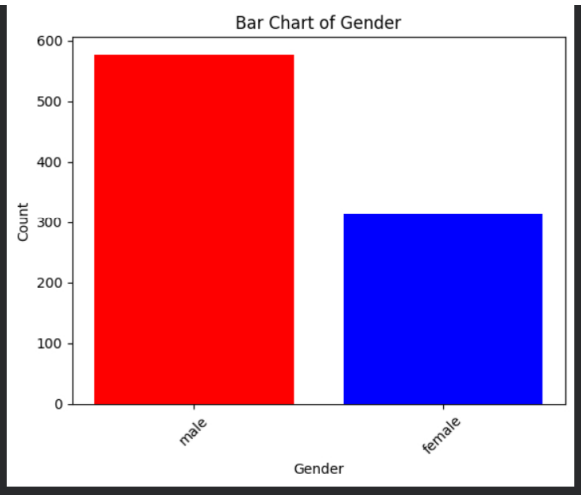
	count
0	577
1	314

dtype: int64

[32]
✓ 0s

```
plt.bar(gender1["sex"], gender1["count"] , color=["red", "blue"])
plt.title("Bar Chart of Gender")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.xticks(rotation = 45)
```

```
([0, 1], [Text(0, 0, 'male'), Text(1, 0, 'female')])
```



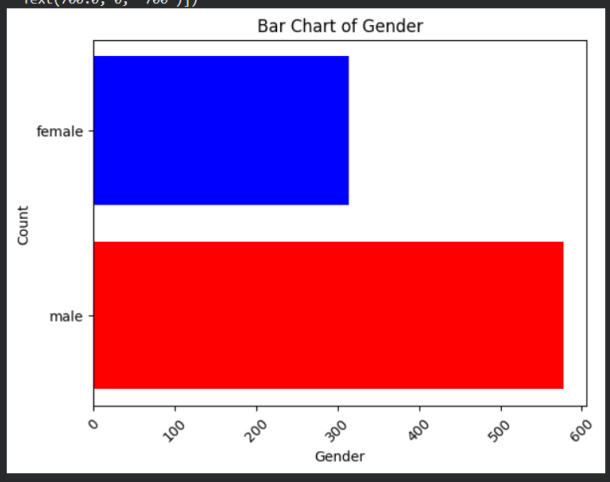
[33] ✓ 0s

```
# horizontal bar chart
```

[35] ✓ 0s

```
plt.barh(gender1["sex"], gender1["count"] , color=["red", "blue"])
plt.title("Bar Chart of Gender")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.xticks(rotation = 45)
```

```
(array([  0., 100., 200., 300., 400., 500., 600., 700.]),
 [Text(0.0, 0, '0'),
  Text(100.0, 0, '100'),
  Text(200.0, 0, '200'),
  Text(300.0, 0, '300'),
  Text(400.0, 0, '400'),
  Text(500.0, 0, '500'),
  Text(600.0, 0, '600'),
  Text(700.0, 0, '700')])
```



[37] ✓ 0s

```
# stacked bar chart
# two category & 1 numerical

# gender wise survival count
```

[38] ✓ 0s

```
# very very important - group by
```

[39] ✓ 0s

```
df.groupby(["sex", "survived"]).size()
```

		0
sex	survived	
female	0	81
	1	233
male	0	468
	1	109

dtype: int64

[41] ✓ 0s

```
gender_survival = df.groupby(["sex", "survived"]).size().unstack()
```

[42] ✓ 0s

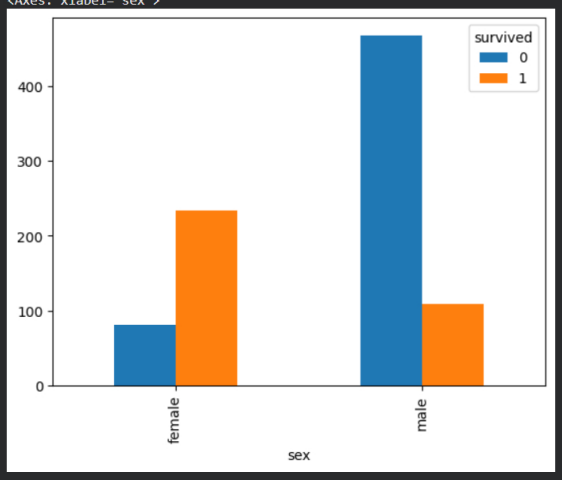
```
gender_survival
```

survived	0	1
sex		
female	81	233
male	468	109

Next steps: Generate code with gender_survival | New interactive sheet

[43]
```
gender_survival.plot(kind = "bar")
```

```
<Axes: xlabel='sex'>
```



[44]
```
# Pie chart - pie chart only in matplotlib library
# pie chart is not in seaborn library
```

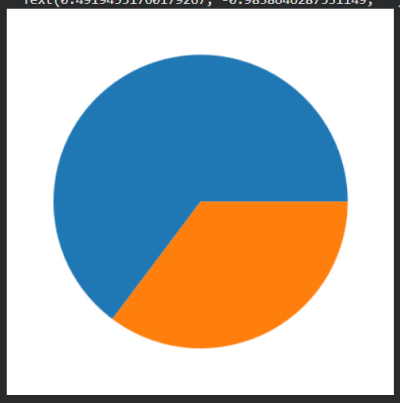
[45]
```
gender
```

sex	count
male	577
female	314

dtype: int64

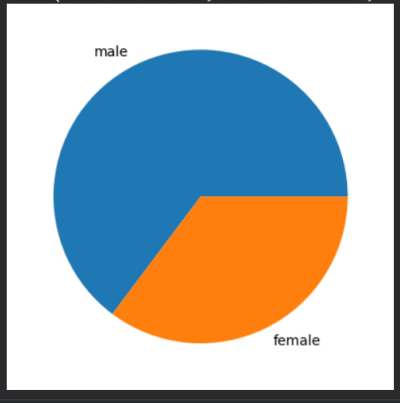
[46]
```
plt.pie(gender.values)
```

```
([<matplotlib.patches.Wedge at 0x79657ef1e120>,
  <matplotlib.patches.Wedge at 0x79657c232210>],
 [Text(-0.4919454315896098, 0.9838646717603559, ''),
  Text(0.4919455176017926, -0.9838646287531149, '')])
```



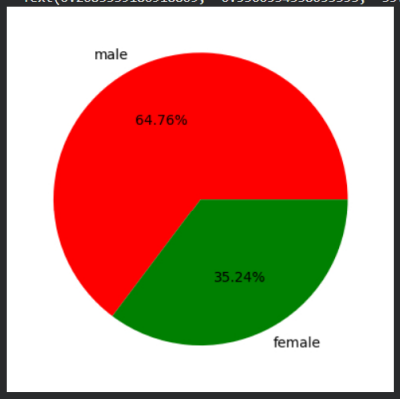
[47]
```
plt.pie(gender.values , labels= gender.index)
```

```
([<matplotlib.patches.Wedge at 0x79657c31bb10>,
  <matplotlib.patches.Wedge at 0x79657c31b890>],
 [Text(-0.4919454315896098, 0.9838646717603559, 'male'),
  Text(0.4919455176017926, -0.9838646287531149, 'female')])
```



[52]
```
plt.pie(gender.values , labels= gender.index,
        autopct="%1.2f%%", colors = ["red", "green"])
```

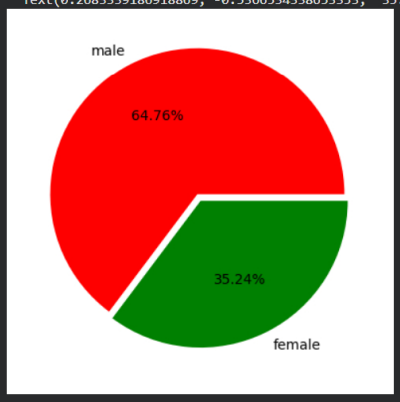
```
([<matplotlib.patches.Wedge at 0x79657c0f1090>,
  <matplotlib.patches.Wedge at 0x79657c0f1450>],
 [Text(-0.4919454315896098, 0.9838646717603559, 'male'),
  Text(0.49194551760179267, -0.9838646287531149, 'female')],
 [Text(-0.2683338717717615076, 0.5366534573238304, '64.76%'),
  Text(0.2683339186918869, -0.5366534338653353, '35.24%')])
```



[54]
✓ 0s

```
plt.pie(gender.values , labels= gender.index,
        autopct="%1.2f%%", explode= [0.05 , 0] ,
        colors = ["red", "green"])
```

```
([<matplotlib.patches.Wedge at 0x79657cd5e990>,
  <matplotlib.patches.Wedge at 0x79657c539310>],
 [Text(-0.5143065875709557, 1.0285857932040083, 'male'),
  Text(0.49194551760179267, -0.9838646287531149, 'female')],
 [Text(-0.29069502775749667, 0.5813745787674829, '64.76%'),
  Text(0.2683339186918869, -0.5366534338653353, '35.24%')])
```



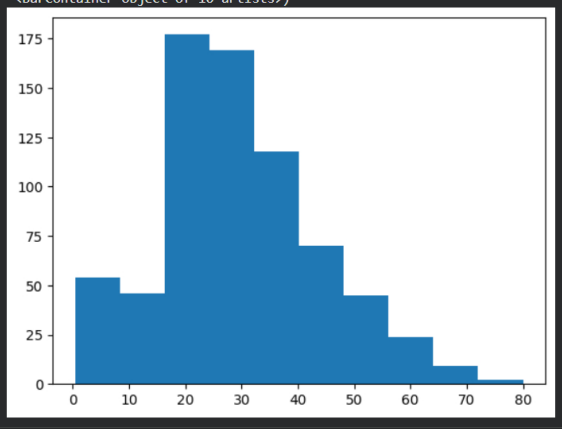
[55]
✓ 0s

```
# histogram  - distribution of numerical data
```

[56]
✓ 0s

```
plt.hist(df["age"])
```

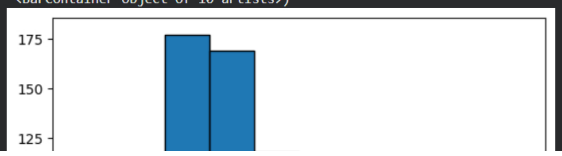
```
(array([ 54.,  46., 177., 169., 118.,  70.,  45.,  24.,   9.,   2.]),
 array([ 0.42 ,  8.378, 16.336, 24.294, 32.252, 40.21 , 48.168, 56.126,
        64.084, 72.042, 80.   ]),
 <BarContainer object of 10 artists>)
```

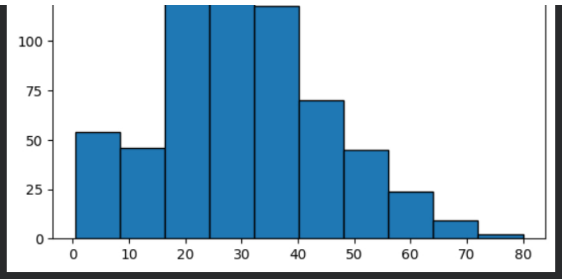


[57]
✓ 0s

```
plt.hist(df["age"], edgecolor ="black")
```

```
(array([ 54.,  46., 177., 169., 118.,  70.,  45.,  24.,   9.,   2.]),
 array([ 0.42 ,  8.378, 16.336, 24.294, 32.252, 40.21 , 48.168, 56.126,
        64.084, 72.042, 80.   ]),
 <BarContainer object of 10 artists>)
```

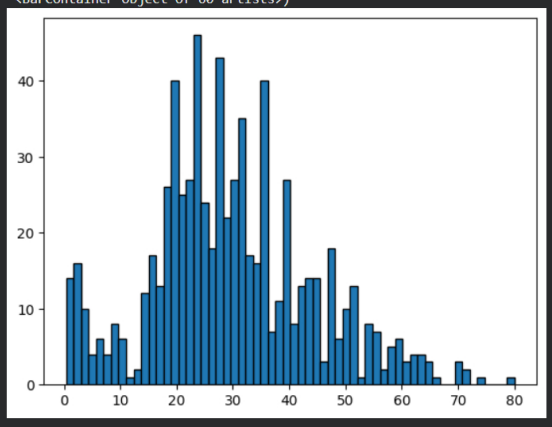




[58]
0s

```
plt.hist(df["age"], edgecolor ="black", bins=60)
```

```
(array([14., 16., 10.,  4.,  6.,  4.,  8.,  6.,  1.,  2., 12., 17., 13.,
        26., 40., 25., 27., 46., 24., 18., 43., 22., 27., 35., 17., 16.,
        40.,  7., 11., 27.,  8., 13., 14., 14.,  3., 18.,  6., 10., 13.,
         1.,  8.,  7.,  2.,  5.,  6.,  3.,  4.,  4.,  3.,  1.,  0.,  0.,
         3.,  2.,  0.,  1.,  0.,  0.,  0.,  1.]),
 array([ 0.42      ,  1.74633333,  3.07266667,  4.399     ,  5.72533333,
         7.05166667,  8.378     ,  9.70433333, 11.03066667, 12.357     ,
        13.68333333, 15.00966667, 16.336     , 17.66233333, 18.98866667,
        20.315     , 21.64133333, 22.96766667, 24.294     , 25.62033333,
        26.94666667, 28.273     , 29.59933333, 30.92566667, 32.252     ,
        33.57833333, 34.90466667, 36.231     , 37.55733333, 38.88366667,
        40.21      , 41.53633333, 42.86266667, 44.189     , 45.51533333,
        46.84166667, 48.168     , 49.49433333, 50.82066667, 52.147     ,
        53.47333333, 54.79966667, 56.126     , 57.45233333, 58.77866667,
        60.105     , 61.43133333, 62.75766667, 64.084     , 65.41033333,
        66.73666667, 68.063     , 69.38933333, 70.71566667, 72.042     ,
        73.36833333, 74.69466667, 76.021     , 77.34733333, 78.67366667,
        80.        ]),
 <BarContainer object of 60 artists>)
```



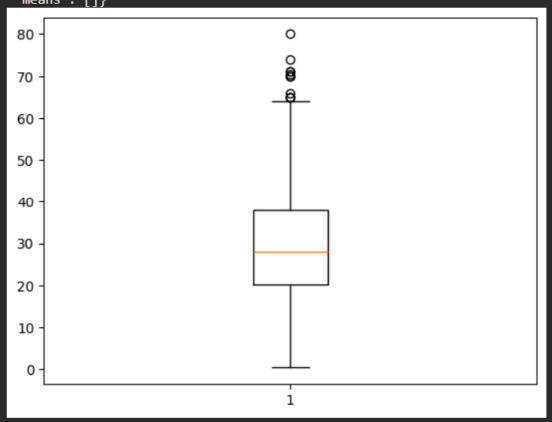
[59]
0s

```
# Outlier detection - BOX plot -
# 5 number summery
```

[61]
0s

```
plt.boxplot(df["age"].dropna())
```

```
{'whiskers': [<matplotlib.lines.Line2D at 0x7965747cc190>,
  <matplotlib.lines.Line2D at 0x7965747cc2d0>],
 'caps': [<matplotlib.lines.Line2D at 0x7965747cc410>,
  <matplotlib.lines.Line2D at 0x7965747cc550>],
 'boxes': [<matplotlib.lines.Line2D at 0x7965747cc050>],
 'medians': [<matplotlib.lines.Line2D at 0x7965747cc690>],
 'fliers': [<matplotlib.lines.Line2D at 0x7965747cc7d0>],
 'means': []}
```



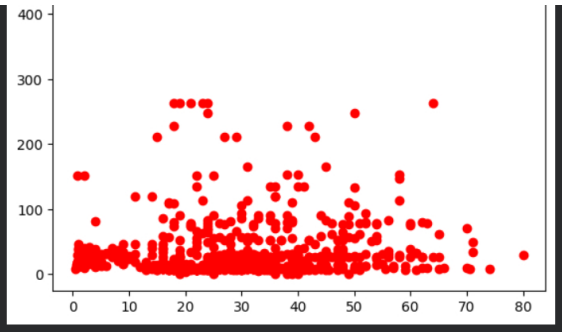
[62]
0s

```
# scatter - numerical  vs numerical
# age - fare
```

[65]
0s

```
plt.scatter(df["age"], df["fare"], color = "red")
plt.savefig("scatterplot.jpeg")
```





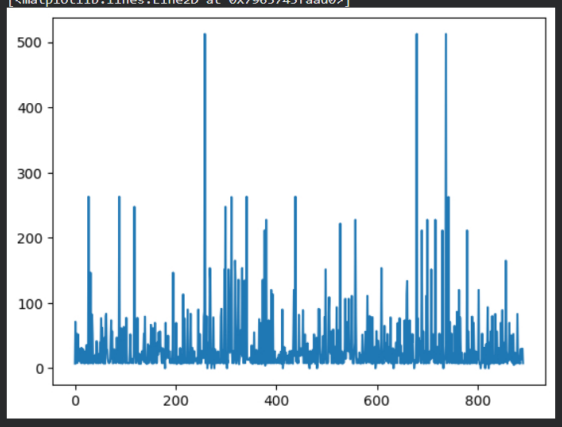
[66] ✓ 0s

```
# line chart
```

[67] ✓ 0s

```
plt.plot(df["fare"])
```

```
[<matplotlib.lines.Line2D at 0x7965745faad0>]
```



[ ]

```
Start coding or generate with AI.
```

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